

Midterm 1 - Solutions

QUESTION 1

Encode flavors availability as a binary triple vcs , where $v = 1$ if vanilla is still available and $v = 0$ otherwise, $c = 1$ if chocolate is still available and $c = 0$ otherwise, $s = 1$ if strawberry is available and $s = 0$ otherwise. Denote A_{vcs} the event that flavors availability is encoded by a triple vcs (for example, A_{001} is the event that vanilla and chocolate are sold out but strawberry is available).

- a) By independence and since all flavors are sold out / available with probability $1/2$, for any $vcs \in \{0, 1\}^3$ we have $\mathbb{P}\{A_{vcs}\} = (1/2)^3 = 1/8$. In particular, $\mathbb{P}\{A_{000}\} = 1/8$.
- b) First, the probability of the condition is $\mathbb{P}\{A_{000}^c\} = 1 - 1/8 = 7/8$, where the upper index c denotes the complement. Next, let P denote the random variable which is the price you will pay for the cheapest cone, and let $\mathbf{1}\{E\}$ be the indicator of event E .

$$\begin{aligned}\mathbb{E}[P\mathbf{1}\{A_{000}^c\}] &= \mathbb{E}[P\mathbf{1}\{A_{001}\}] + \mathbb{E}[P\mathbf{1}\{A_{010}\}] + \mathbb{E}[P\mathbf{1}\{A_{011}\}] + \\ &\quad + \mathbb{E}[P\mathbf{1}\{A_{100}\}] + \mathbb{E}[P\mathbf{1}\{A_{101}\}] + \mathbb{E}[P\mathbf{1}\{A_{110}\}] + \mathbb{E}[P\mathbf{1}\{A_{111}\}] = \\ &= 4\mathbb{E}[\mathbf{1}\{A_{001}\}] + 2\mathbb{E}[\mathbf{1}\{A_{010}\}] + 2\mathbb{E}[\mathbf{1}\{A_{011}\}] + \\ &\quad + 1\mathbb{E}[\mathbf{1}\{A_{100}\}] + 1\mathbb{E}[\mathbf{1}\{A_{101}\}] + 1\mathbb{E}[\mathbf{1}\{A_{110}\}] + 1\mathbb{E}[\mathbf{1}\{A_{111}\}] = \frac{12}{8},\end{aligned}$$

where we used $\mathbb{E}[\mathbf{1}\{E\}] = \mathbb{P}\{E\}$ for any event E and $\mathbb{P}\{A_{vcs}\} = 1/8$ for any $vcs \in \{0, 1\}^3$. Finally,

$$\mathbb{E}[P|A_{000}^c] = \frac{\mathbb{E}[P\mathbf{1}\{A_{000}^c\}]}{\mathbb{P}\{A_{000}^c\}} = \frac{12/8}{7/8} = \frac{12}{7}.$$

- c) The event that you do not buy an ice cream cone is $A_{000} \cup A_{001}$. By the definition of conditional probability,

$$\mathbb{P}\{A_{000}|A_{000} \cup A_{001}\} = \frac{\mathbb{P}\{A_{000}\}}{\mathbb{P}\{A_{000} \cup A_{001}\}} = \frac{1/8}{1/8 + 1/8} = \frac{1}{2}.$$

Grading comments:

- One very common mistake in part b) was to calculate $\mathbb{E}[p\mathbf{1}\{A_{000}^c\}]$ and not $\mathbb{E}[p|A_{000}^c]$.
- Students sometimes illegally used $\mathbf{1}\{A \cup B\} = \mathbf{1}\{A\} + \mathbf{1}\{B\}$ for non-disjoint events A and B (or the law of total expectation with non-disjoint events, which is equivalent in this context).
- Students sometimes misunderstood how conditioning works: when writing the sum $4\mathbb{P}\{A_{001}\} + 2\mathbb{P}\{A_{010}\} + \dots$, they assumed that $\mathbb{P}\{A_{001}\}$ now equals $1/4$ and not $1/8$ since if vanilla and chocolate are unavailable, strawberry has to be available because of conditioning. Actually $\mathbb{P}\{A_{001}|A_{000}^c\} = \mathbb{P}\{A_{001}\}/\mathbb{P}\{A_{000}^c\} = 1/7$ and this concerns all the terms of the sum, not just the 001 term.
- Students commonly applied the Bayes rule in part c) and then used a wrong number for one of the resulting probabilities in the fraction.

QUESTION 2

- a) Let X be the number of disgruntled students $\Rightarrow X \sim \text{Bin}(100, 0.01)$. Hence, the probability of exactly $\frac{1}{4}$ of the class signs the petition is:

$$\mathbb{P}(X = 25) = \binom{100}{25} 0.01^{25} 0.99^{75}.$$

- b) Let F be the event the professor gets fired and $\tilde{X}$ be the fraction of students that sign the petition. Note, $\tilde{X} = \frac{X}{100} \in \{\frac{0}{100}, \frac{1}{100}, \dots, \frac{100}{100}\}$.

Hence, we have:

$$\mathbb{P}(F) = \mathbb{P}\left(\bigcup_{k=0}^{100} (F \cap \{\tilde{X} = \frac{k}{100}\})\right) \quad (1)$$

$$= \sum_{k=0}^{100} \mathbb{P}(F \cap \{\tilde{X} = \frac{k}{100}\}) \quad (2)$$

$$= \sum_{k=0}^{100} \mathbb{P}(F \cap \{X = k\}) = \sum_{k=0}^{100} \mathbb{P}(F|X = k) \mathbb{P}(X = k) \quad (3)$$

$$= \sum_{k=0}^{100} \left(\frac{k}{100}\right)^2 \binom{100}{k} 0.01^k 0.99^{100-k} \quad (4)$$

$$= \frac{1}{100^2} \sum_{k=0}^{100} k^2 \binom{100}{k} 0.01^k 0.99^{100-k} \quad (5)$$

$$= \frac{1}{100^2} \mathbb{E}(X^2) \quad (6)$$

$$= \frac{1}{100^2} ((\mathbb{E}(X))^2 + \text{Var}(X)) \quad (7)$$

$$= \frac{1}{100^2} ((100 \times 0.01)^2 + 100 \times 0.01 \times 0.99) = 1.99 \cdot 10^{-4} \quad (8)$$

Grading comments:

- (a): Almost all students did part a correctly
- (b): Very few students were able to complete part b.
- Most students were not able to simplify the algebra and stopped at equation 5 (written above) as it seems like they could not observe that the sum is the second moment of X , i.e, $E(X^2)$.
- A lot of students mistook $E(X^2)$ for $[E(X)]^2$.
- Similarly, a good number of students claimed that $\mathbb{P}(F) = \left(\sum_{k=0}^{100} \left(\frac{k}{100}\right) \binom{100}{k} 0.01^k 0.99^{100-k}\right)^2$ instead of $\sum_{k=0}^{100} \left(\frac{k}{100}\right)^2 \binom{100}{k} 0.01^k 0.99^{100-k}$.

QUESTION 3

a) Denote by S_1, S_2 the numbers of sprinkles on the two cupcakes.

Then we have that:

(1) S_1 and S_2 are independent

(2) $P(S_i = n) = p(1 - p)^{n-1}$ for $n \geq 1, i = 1, 2$.

Note that

$$\begin{aligned}\{S_1 = S_2\} &= \cup_{n=1}^{\infty} \{S_1 = S_2 = n\} \\ &= \cup_{n=1}^{\infty} (\{S_1 = n\} \cap \{S_2 = n\})\end{aligned}$$

So we have

$$\begin{aligned}P(S_1 = S_2) &= \sum_{n=1}^{\infty} P(\{S_1 = n\} \cap \{S_2 = n\}) \text{ by the third principle, the events are disjoint} \\ &= \sum_{n=1}^{\infty} P(S_1 = n)P(S_2 = n) \text{ by independence} \\ &= \sum_{n=1}^{\infty} p(1 - p)^{n-1} \cdot p(1 - p)^{n-1} \\ &= p^2 \sum_{n=1}^{\infty} ((1 - p)^2)^{n-1} \\ &= p^2 \frac{1}{1 - (1 - p)^2} \text{ by geometric series formula} \\ &= \frac{p}{2 - p}.\end{aligned}$$

b) $S_1 + S_2 \geq 1 + 1 = 2$. For any integer $k \geq 2$, we have

$$\begin{aligned}\{S_1 + S_2 = k\} &= \cup_{n=1}^{k-1} \{S_1 = n, S_2 = k - n\} \\ &= \cup_{n=1}^{\infty} (\{S_1 = n\} \cap \{S_2 = k - n\})\end{aligned}$$

So we have

$$\begin{aligned}P(S_1 + S_2 = k) &= \sum_{n=1}^{k-1} P(\{S_1 = n\} \cap \{S_2 = k - n\}) \\ &= \sum_{n=1}^{k-1} P(S_1 = n)P(S_2 = k - n) \text{ by independence} \\ &= \sum_{n=1}^{k-1} p(1 - p)^{n-1} \cdot p(1 - p)^{k-n-1} \\ &= p^2 \sum_{n=1}^{k-1} (1 - p)^{k-2} = p^2(k - 1)(1 - p)^{k-2}.\end{aligned}$$

The range of $S_1 + S_2$ is $\{k \in \mathbb{N} | k \geq 2\}$. So the distribution of $S_1 + S_2$ is given by

$$P(S_1 + S_2 = k) = p^2(k - 1)(1 - p)^{k-2} \text{ for } k = 2, 3, 4, \dots$$

Grading comments:

- The common mistakes in (a) include:
 1. wrongly interpreting the event $\{S_1 = S_2\}$ as the event $\{S_1 = S_2 = n\}$ for a single specific n .
 2. wrongly specifying the summation index for $P(S_1 = S_2) = p^2 \sum_{n=1}^{\infty} ((1-p)^2)^{n-1}$. The index starts from $n = 1$, not $n = 0$, because Geometric distributed random variable $S_1, S_2 \geq 1$.
 3. calculation error about geometric series summation.
- The common mistakes in (b) include:
 1. wrongly interpreting the event $\{S_1 + S_2 = k\}$
 2. not pointing out that the range of the random variable $(S_1 + S_2)$ is $\{k \in \mathbb{N} | k \geq 2\}$. $k \geq 2$ is a very important constraint.
 3. calculation error about $P(S_1 + S_2 = k) = \sum_{n=1}^{k-1} p(1-p)^{n-1} \cdot p(1-p)^{k-n-1}$.
 4. directly using the conclusion that “sum of two Geometric random variable is Negative-Binomial distribution” without any proof.